

~min sheet~

Preservation

(Preservation) If $\vdash e : T$ and $e \mapsto e'$
then $\vdash e' : T$

Proof by induction on $e \mapsto e'$.

[Case: D-LET]

GOAL = If $\vdash \text{let}(e_1; x. e_2) : T$
and $\text{let}(e_1; x. e_2) \mapsto e_2[e_1/x]$
then $\vdash e_2[e_1/x] : T$

$P_1 = \vdash \text{let}(e_1; x. e_2) : T$
 $P_2 = \text{let}(e_1; x. e_2) \mapsto e_2[e_1/x]$

GOAL' = $\vdash e_2[e_1/x] : T$

By inversion on P_1 , we know that there must exist a type σ such that

- $\vdash e_1 : \sigma$ (INV1)
- $x : \sigma \vdash e_2 : T$ (INV2)

(Subst) If $\vdash e : T$ and $\vdash x : T \vdash u : \sigma$
then $\vdash u[e/x] : \sigma$

Specialised subst

If $\vdash e_1 : \sigma$ and $x : \sigma \vdash e_2 : T$
then $\vdash e_2[e_1/x] : T$

We can conclude our goal by applying the substitution lemma to INV1 and INV2.

Progress

(using canonical forms from yesterday)

(Progress) If $\vdash e : \tau$ then either
 $e \text{ val}$, or
 $e \mapsto e'$ for some e' .

Proof by induction on $\vdash e : \tau$

[Case: PLUS]

GOAL = If $\vdash \text{plus}(e_1; e_2) : \tau$ then either
 $\text{plus}(e_1; e_2) \text{ val}$ or
 $\text{plus}(e_1; e_2) \mapsto e'$ for some e' .

P1 = $\vdash \text{plus}(e_1; e_2) : \tau$

GOAL' = either $\text{plus}(e_1; e_2) \text{ val}$, or
 $\text{plus}(e_1; e_2) \mapsto e'$ for some e'

We will show the latter because plus is not a value.

IH1 = if $\vdash e_1 : \tau_1$ then either $e_1 \text{ val}$, or
 $e_1 \mapsto e'_1$ for some e'_1

IH2 = if $\vdash e_2 : \tau_2$ then either $e_2 \text{ val}$, or
 $e_2 \mapsto e'_2$ for some e'_2

By inversion on P1 we can conclude:

- $\tau = \text{Num}$ (INV1)
- $\vdash e_1 : \text{Num}$ (INV2)
- $\vdash e_2 : \text{Num}$ (INV3)

By applying IH1 to INV2, we can conclude

UIH1 = either $e_1 \text{ val}$, or
 $e_1 \mapsto e'_1$ for some e'_1

By applying IH2 to INV3, we can conclude

UIH2 = either $e_2 \text{ val}$, or
 $e_2 \mapsto e'_2$ for some e'_2

We proceed by case analysis on UIH1 and UIH2

[Subcase: $e_1 \text{ val}$]

[Subcase: $e_2 \text{ val}$]

By canonical forms, we have

$$e_1 = \text{num}[n_1]$$

$$e_2 = \text{num}[n_2]$$

This means that we can apply the D-PLUS rule to get our e' and achieve our goal.

$$\text{D-PLUS} \frac{n_1 + n_2 = n}{\text{plus}(\text{num}[n_1]; \text{num}[n_2]) \mapsto \text{num}[n]}$$

Thus $e' = \text{num}[n]$

[Subsub: $e_2 \mapsto e'_2$ for some e'_2]

We can achieve our goal via the D-PLUS-2 rule:

$$\text{D-PLUS-2} \frac{\begin{array}{cc} \text{(subcase)} & \text{(subsubcase)} \\ e_1 \text{ val} & e_2 \mapsto e'_2 \end{array}}{\text{plus}(e_1; e_2) \mapsto \text{plus}(e_1; e'_2)}$$

Thus $e' = \text{plus}(e_1; e'_2)$

[Subcase: $e_1 \mapsto e'_1$ for some e'_1]

Regardless of what e_2 is, we can produce e' (goal) using D-PLUS-1

$$\frac{\begin{array}{c} \text{(subcase)} \\ e_1 \mapsto e'_1 \end{array}}{\text{plus}(e_1; e_2) \mapsto \text{plus}(e'_1; e_2)} \text{D-PLUS-1}$$

Thus we have achieved our goal with $e' = \text{plus}(e'_1; e_2)$

We have exhaustively covered all the cases.

Thus we are done for PLUS.

Functions

$$(\lambda x \rightarrow x) :: a \rightarrow a$$

Syntax:

types $\tau ::= \dots$
 $\tau_1 \rightarrow \tau_2$

pre-terms $e ::= \dots$
 $\lambda x:\tau. e$
 $e_1(e_2)$

Statics:

LAM
$$\frac{\Gamma, x:\sigma \vdash e:\tau}{\Gamma \vdash \lambda x:\sigma. e : \sigma \rightarrow \tau}$$

APP
$$\frac{\Gamma \vdash e_1 : \sigma \rightarrow \tau \quad \Gamma \vdash e_2 : \sigma}{\Gamma \vdash e_1(e_2) : \tau}$$

Q What is the difference between let and lam.

A See Teams